The Barn Effect (Tight Exposition)

Pairwise Tabular Interaction Graphs and Row Attributions

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May 7, 2026

Abstract

Given a tabular sample with scalar outcome $Y^{(n)} \in \mathbb{R}$ (label, probability, or model output), we discretize each feature to bins, then accumulate co-occurrence counts C_{uv} and outcome sums W_{uv} over all unordered column pairs (i, j) and encoded bin pairs (u, v). The artifact is a sparse weighted graph $\mathbf{\Phi}$ with $\Phi_{uv} = (W_{uv}/C_{uv}) \log_{10}(C_{uv})$ when $C_{uv} > 1$, else 0. Training and inference are chunk-additive; storage is $O(p^2 B^2)$ in bin depth B. We record a.s. convergence of a scaled Φ_{uv} to the stratum conditional mean under i.i.d. sampling and *fixed* quantization, and note identifiability limits for empty or singleton cells [1, 5]. No experiments; full discussion appears in `barn_effect.tex`.

1 Problem and scope

We target *descriptive* row-level signals from empirical co-occurrence structure, complementary to model-based explainers (e.g. [6, 7]) and to ICE-style curve views [4]. Streaming sufficiency follows [8, 9]. This note states definitions and two propositions only.

2 Preliminaries

Dataset $\mathcal{D} = \{(\boldsymbol{x}^{(n)}, Y^{(n)})\}_{n=1}^{N}$, features $i \in [p]$, values $X_i^{(n)}$. Column pairs $\mathcal{C} = \{(i, j) : 1 \le i < j \le p\}$. Optional chunks $\mathcal{D}_k$ partition $\mathcal{D}$.

3 Phase 1: quantization

Definition 1 (Bin schema)**.** *Fix B_i.* Numerical i: *range $r_i = \max_n X_i^{(n)} - \min_n X_i^{(n)}$; round to s_i decimal places via $s_i = \max(0, 4 - \lfloor \log_{10} r_i \rfloor)$; take top-$B_i$ rounded frequencies as interior split candidates, replace largest candidate by $\max_n X_i^{(n)}$, prepend $\min_n X_i^{(n)}$, form B_i interval bins.* Categorical i: *keep top-B_i levels; map others to ε. Write $\mathcal{B} = \{(\mathcal{L}_i, \ell^{(i)})\}$, $\ell^{(i)} : \mathrm{dom}(X_i) \to \mathcal{L}_i$. Disjoint column encodings identify $\mathcal{V} = \bigcup_i \mathcal{L}_i$ with $[V]$ so cross-column bin pairs are unambiguous.*

4 Phase 2: accumulation and graph

For $(i,j) \in \mathcal{C}$, $i<j$, and each chunk, add to sparse (W,C):

$$\Delta W_{uv}^{(k)} = \sum_{\substack{n\in\mathcal{D}_k:\\ \ell^{(i)}(X_i^{(n)})=u,\,\ell^{(j)}(X_j^{(n)})=v}} Y^{(n)}, \qquad \Delta C_{uv}^{(k)} = \big|\{n \in \mathcal{D}_k : \cdots\}\big|. \tag{1}$$

Summing over chunks gives global W_{uv}, C_{uv} (partition-invariant).

Definition 2 (Interaction graph)**.** $\Phi_{uv} = 0$ *if* $C_{uv} \in \{0,1\}$*, else* $\Phi_{uv} = (W_{uv}/C_{uv})\log_{10}(C_{uv})$*.*

5 Inference

For row n, $\psi_{ij}^{(n)} = \Phi_{\ell^{(i)}(X_i^{(n)}),\ell^{(j)}(X_j^{(n)})}$. Let $d_c^{(n)} = |\{j \neq c : \psi_{cj}^{(n)} \neq 0\}|$.

$$A_c^{(n)} = \begin{cases} 0 & d_c^{(n)} = 0, \\ \dfrac{1}{d_c^{(n)}} \sum_{j\neq c} \psi_{cj}^{(n)} & \text{else.} \end{cases} \qquad \delta_c^{(n)} = A_c^{(n)} - \bar{Y}, \quad \bar{Y} = \frac{1}{N}\sum_n Y^{(n)}.$$

Global score $\bar{\delta}_c = \frac{1}{N}\sum_n \delta_c^{(n)}$. Attributions do *not* satisfy an efficiency axiom $\sum_c \delta_c^{(n)} = Y^{(n)} - \bar{Y}$ [6].

6 Properties

Remark 1. *Chunks only change how* (1) *is scheduled; totals match one pass over* $\mathcal{D}$*.*

Proposition 1 (Complexity)**.** *Training:* $O(p^2N)$ *time,* $O(p^2B^2)$ *sparse keys. Inference per row:* $O(p^2)$ *lookups.*

Remark 2 (Fixed $\mathcal{B}$ for limits)**.** *Proposition 2 treats* $\mathcal{B}, \ell^{(i)}$ *as nonrandom; Definition 1 instantiates empirical* $\mathcal{B}_N$ *on the sample. Joint limits for* $(\mathcal{B}_N, \mathbf{\Phi})$ *are not pursued.*

Proposition 2 (Scaled a.s. limit for Φ_{uv})**.** *If* $(\boldsymbol{x}^{(n)}, Y^{(n)})$ *are i.i.d.,* $\mathbb{E}|Y| < \infty$*,* $\mathcal{B}$ *fixed, and* $\pi = P(\ell^{(i)}(X_i) = u, \ell^{(j)}(X_j) = v) > 0$*, then*

$$\frac{\Phi_{uv}}{\log_{10}(N\pi)} \xrightarrow{\text{a.s.}} \mathbb{E}\big[Y \mid \ell^{(i)}(X_i) = u, \ell^{(j)}(X_j) = v\big].$$

Sketch. $C_{uv}/N \to \pi$ a.s.; $\log_{10} C_{uv} - \log_{10} N \to \log_{10}\pi$ a.s.; $W_{uv}/C_{uv} \to \mathbb{E}[Y \mid u, v]$ on the stratum. Combine and rescale. □

Remark 3 (Support)**.** *Zero* Φ_{uv} *does not imply* $\mathbb{E}[Y \mid u,v] = 0$*:* $C_{uv} = 0$ *means unseen;* $C_{uv} = 1$ *is zeroed by construction. Sparse cells — partial identification [5]; inference cautions [1]. The divisor* $d_c^{(n)}$ *averages only partners with nonzero* Φ *support at inference (data-relative context), not a full* $p-1$ *counterfactual split.*

7 Implementation (minimal)

Realize (W,C) as one hash map on canonical (u,v) with $i<j$. $Y^{(n)}$ may be raw y, class probability, or $\hat{f}(\boldsymbol{x}^{(n)})$ (same pipeline). Typical engineering: column-prefixed string keys for $\mathcal{V}$, NaN handling fixed *before* binning; recompute $\mathbf{\Phi}$ from (W,C) after all chunks.

8 Related work (one pass)

Co-occurrence of discretized items parallels frequent-pattern mining [2] but W supplies a supervised conditional mean, not binary support alone. The $\log_{10} C$ factor resembles frequency dampers in IR [3].

Conclusion

The Barn Effect is a discrete pairwise sufficient-statistic summary $\mathbf{\Phi}$ plus row scores $\mathbf{\Delta}$, streamable and $O(p^2 B^2)$-bounded. The main theoretical loose end is extending Proposition 2 to empirical $\mathcal{B}_N$ (Remark 2).

References

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